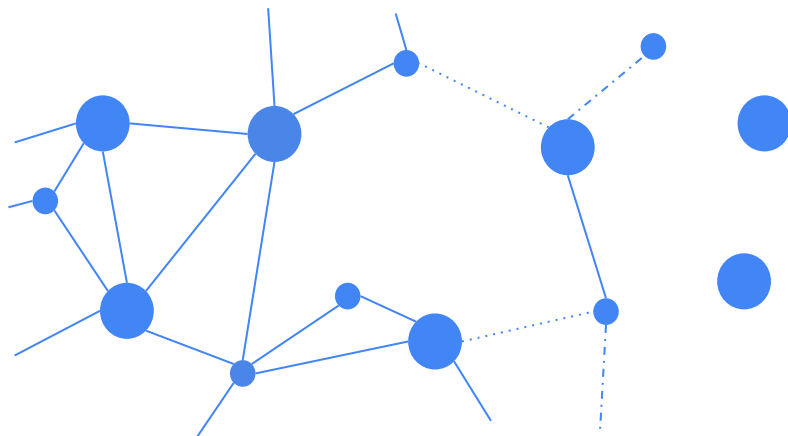


Making cultural heritage collections usable as data: challenges, limitations and opportunities

Gustavo Candela

10th September 2026



About me

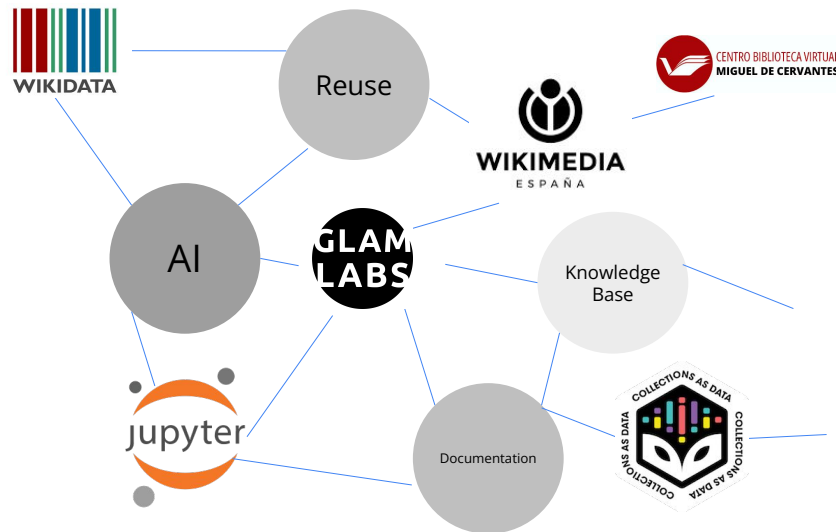
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- Developer [Biblioteca Virtual Miguel de Cervantes](#) 2010-2023
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 - [International GLAM Labs Community](#)
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 - [Technical Assessment Committee Echoes](#)
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Content

- Introduction
- Challenges & opportunities



My sincere thanks to all the researchers I have had the pleasure of collaborating with, whose invaluable contributions, inspiring ideas, and generosity have enriched my research journey.

Introduction

1. Collections as data



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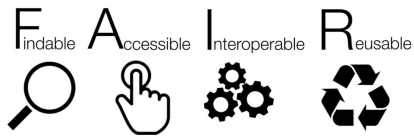
Universitat d'Alacant
Universidad de Alicante



DARIAH-EU



Collections as Data



Focused on data



Complement FAIR

<https://collectionsasdata.github.io/>

<https://www.gida-global.org/care>

Findable

Rich metadata and permanent identifiers

Accessible

Machine-readable metadata

Interoperable

Metadata described with vocabularies

Reusable

Licence and documentation

Collective benefit

Inclusive and equitable outcomes

Authority to control

Recognizing rights and interests

Responsibility

For indigenous languages and worldviews

Ethics

Minimizing harm, and for justice and future use

A museum digitises Indigenous artifacts, photographs, or recordings.

Applying CARE could mean that Indigenous communities help decide:

- what gets digitised;
- what remains restricted;
- how objects and knowledge are described;
- who can access the digital collection; and
- whether materials can be used commercially.

Publishing Collections as Data

Publication

A checklist to make available digital collections suitable for computational use



<https://doi.org/10.48550/arXiv.2304.02603>

[Global Knowledge, Memory and Communication](#)

☒ A Checklist to Publish Collections as Data in GLAM Institutions

☐ 01	Provide a clear license allowing reuse of the dataset without restrictions	• CC0, CC BY, Public Domain Mark • National initiatives • No known copyright
☐ 02	Provide a suggestion of how to cite the dataset	• BibTeX, APA • DOI • Versions
☐ 03	Include documentation about the dataset	• README file • Tutorials and websites • Programming Historian & research articles
☐ 04	Use a public platform to publish the dataset	• GitHub, Zenodo, DataCite • Hosting
☐ 05	Share examples of use as additional documentation	• Prototypes & tools • Jupyter Notebooks (reproducible) • GLAM Labs
☐ 06	Give structure to the dataset	• Folder structure • Using self-describing folder and file names • BagIt File Packaging Format & Data Package
☐ 07	Provide machine-readable metadata (about the dataset itself)	• Dublin Core • Vocabulary of Interlinked Datasets (VoID) • Data Catalog Vocabulary (DCAT)
☐ 08	Include your dataset in collaborative edition platforms	• Increase visibility • Title, author, location, license, main subject, etc.
☐ 09	Offer an API access to your repository	• OAI-PMH, JSON, XML • IIIF • SPARQL
☐ 10	Develop a portal page	• GitHub Pages • New section in the Lab • e.g. Chronicling America & Data Foundry
☐ 11	Add a terms of use	• e.g. section detailing copyright, liability and access statements

<https://www.glamlabs.io/checklist>



DARIAH-EU

The Digital Research Infrastructure for Arts and Humanities supports digitally-enabled research and teaching.



CLARIN

Provides access to multimodal digital language data and tools with which to explore, analyse or combine these datasets.



European Open Science Cloud

A platform for hosting and processing research data to support EU science.



SSHOC

The SSHOC Cluster brings together 20 research organisations that play a key role in Social Sciences and Humanities.

Collections as Data

Advances in technology such as artificial intelligence and machine learning have provided a new context in which Collections as Data can be analysed using computational methods.



The workflow

A step by step guide for cultural heritage institutions for publishing their collections so that they are suitable for computational use.

[ACCESS THE WORKFLOW](#)


The article

The accompanying article A Workflow to Publish Collections as Data provides additional context for the concept.

[READ THE ARTICLE](#)


The course

Find the DARIAH-Campus resource introduction to Collections as Data to learn how to publish them.

[ACCESS THE COURSE](#)

Challenges

1. What is a digital collection as Collections as data?
2. Copyright
3. Quality
4. Knowledge description
5. Uneven readiness for institutions



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What is a digital collection as Collections as data?



- A digital collection in Collections as Data is not simply a set of digitized objects online; it is a collection that can be treated as a **dataset for computational research and reuse**
- Potential content
 - Metadata, content, OCR, text, images, newspapers, maps, etc.
 - Detailed documentation about the digital collection
 - License and terms of use
 - Reproducible code
 - Examples of use and prototypes
 - References
 - ...



<https://www.dataspace-culturalheritage.eu/en/news/what-is-a-dataset-share-your-thoughts>

Copyright

- **Clear licenses to promote the reuse of the digital collections**
- “Openly available online” does not necessarily mean “free to reuse as data”
 - For example, a library might put 100,000 digitized photographs online. Users can view them through the library website, but copyright might prevent users from downloading, reproducing, or computationally analyzing those photographs.
- **Federated/secure training:** For some AI projects, data can remain inside the institution while models learn from it. This can reduce copying and data-transfer risks, but it does not itself solve copyright authorization.



*More about legal issues
with Andrea Wallace ...*

Quality

Dişli, M., Candela, G., Gutiérrez, S. and Fontenelle, G. (2025) 'Open Data Practices of Art Museums in Wikidata: A Compliance Assessment', *Journal of Open Humanities Data*, 11(1), p. 71. <https://doi.org/10.5334/johd.438>

- **Metadata** quality
 - Data quality criteria: completeness, accuracy, understandability, etc.
- **Optical Character Recognition (OCR)** quality
 - Text reviewed
- **Enrichment**
 - External repositories: Wikidata
 - Data silos
- **Documentation & examples of use**
 - Detailed description of the curation process, content, biases, examples of use, etc.
 - Required time to write the documentation?
 - Machine-readable documentation?
 - Examples of use: Notebooks and prototypes



Knowledge description - Ontologies

- **Difficult to understand for non-expert users**
- Complex workflow to maintain (e.g., triple storage systems)
- Ontologies cover, **to some extent**, the description of knowledge
- Different institutions use different ontologies and controlled vocabularies to describe their catalogues, leading to **data silos**

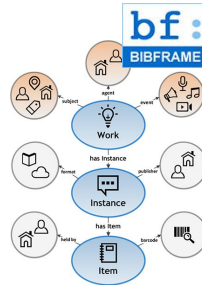
Classes and
properties to
describe the content



[Europeana Data Model](https://www.europeana.eu/)

Schema.org

<https://schema.org/>



<https://www.loc.gov/bibframe/>



<https://cidoc-crm.org/>



International
Federation of
Library
Associations and Institutions

[IFLA Library Rerefence Model](https://www.ifla.org/)



<https://www.wikidata.org/>

Exploring the use of SPARQL federated queries

Avoiding data silos ...



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From Linked Open Data to Collections as Data

A Reproducible Framework Using Federated Queries

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DOI: <https://doi.org/10.5860/ital.v44i4.17432>

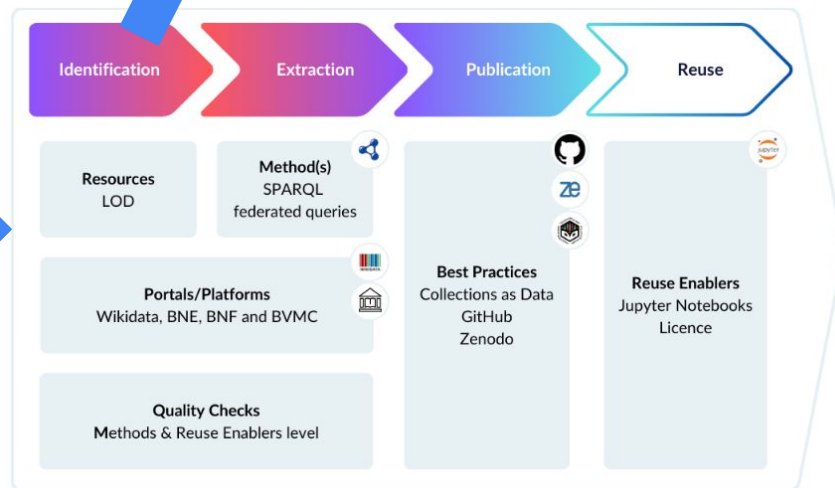
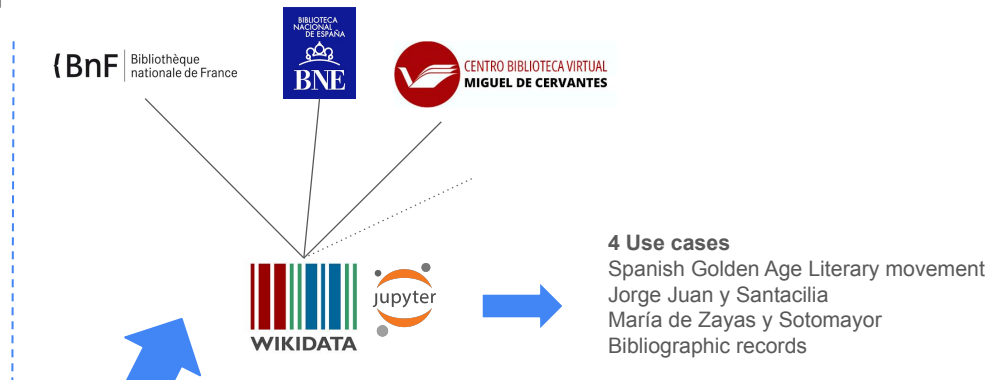
PDF

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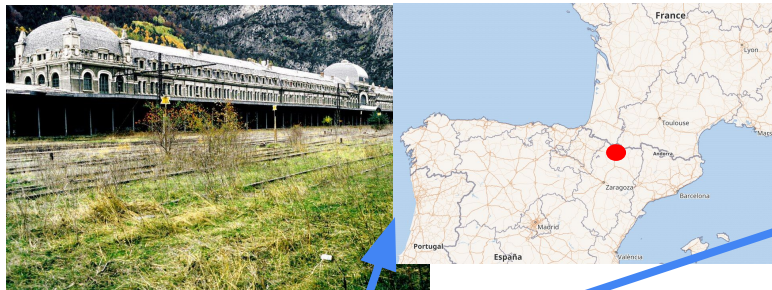
Knowledge description - Ontologies

Existing ontologies and vocabularies enable the description of any kind of information? to what extent? the combination of several of them could help?

- **Example 1:** How do we map metadata from Dublin Core to CIDOC? Is any information missing in the mapping process? 15 metadata fields vs CIDOC CRM 7.1.3 has 160 properties and 81 classes
- **Example 2:** Challenges when transforming traditional MARC records to ontologies based on several layers Work-Expression-Manifestation-Item (FRBR)?
- **There is no single ontology capable of describing every conceivable kind of information**, but the existing ecosystem provides a very large and extensible set of semantic building blocks from which complex descriptions can be constructed

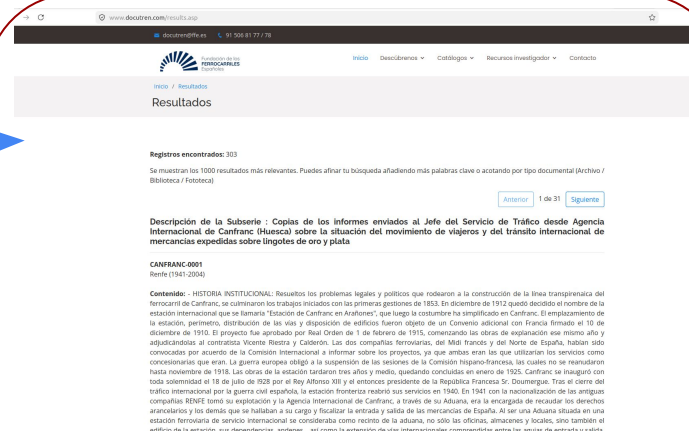


Canfranc International Railway Station - WWII

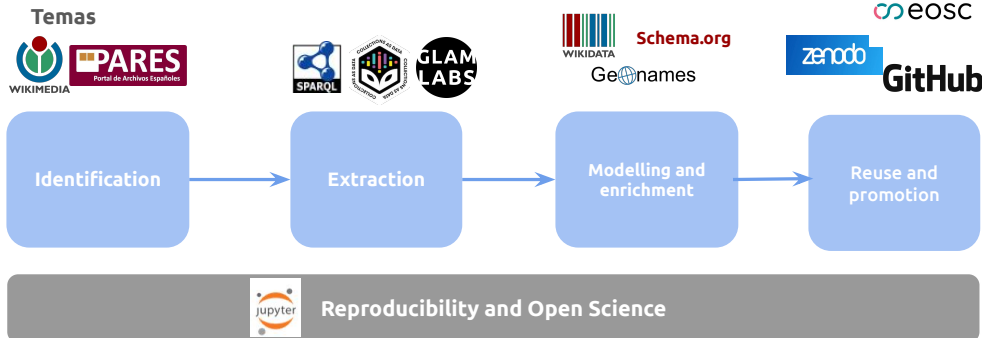


Ontological model to describe events

Albert Le Lay, a customs officer, helped refugees, the Allies, and the French Resistance.



Nazi gold in exchange for tungsten.



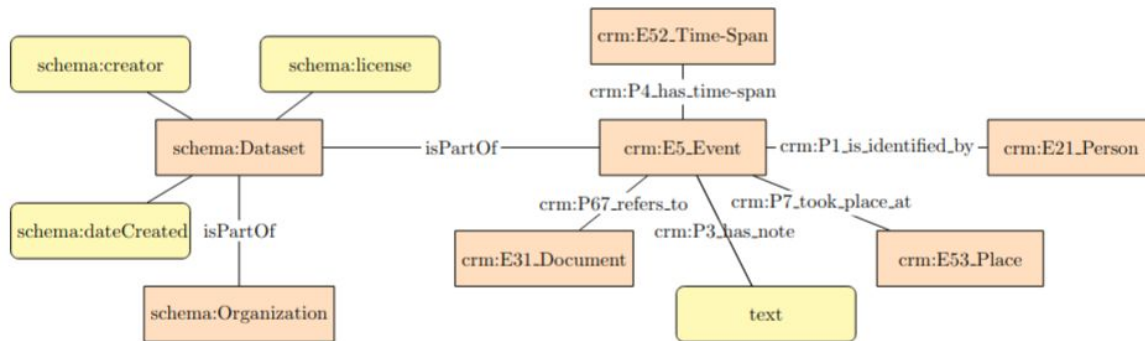
Thanks to Paul Spence

Canfranc International Railway Station - WWII



Schema.org

Ontological model to describe events



Events

- Albert Le Lay as head of the French customs office in Canfranc
- Transport of information and liaison with Allied resistance and intelligence networks
- Documented movements of gold at the station
- Commercial activity

<https://github.com/hibernator11/canfranc-ww2>

Thanks to Paul Spence

Uneven readiness for institutions

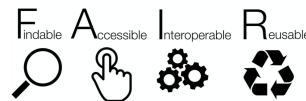
- **Traditional work VS new requirements**
 - New jobs/skills: data science
- **Limited resources**
 - Staff, time and funding
- Promoting **training materials** and courses:
 - [DARIAH Campus](#)
 - [Digital Scholarship & Data Science Topic Guides for Library Professionals \(DS Topic Guides\)](#)
- **AI Literacy**
 - Maintaining relevance in GLAM institutions

*More about courses and
Open Research with
Nadezhda Povroznik ...*



AI4LAM

Artificial Intelligence for Libraries, Archives & Museums

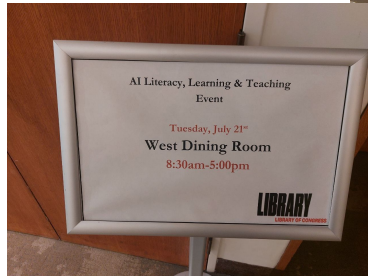


CARE Principles
for Indigenous
Data Governance



Opportunities

- **AI Literacy** in GLAM: communities
- **Interdisciplinary research:** GLAM staff, engineers, mathematicians, historians, linguists, etc.
 - Data infrastructures
- *“As **open** as possible and as closed as necessary”*
- **Documentation** for transparent use of the data
- Defining a **minimal AI infrastructure** for a local environment?



AI4LAM

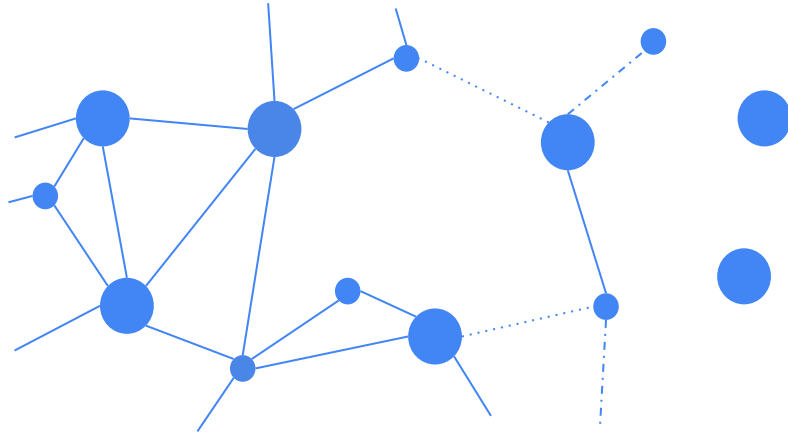
Artificial Intelligence for Libraries, Archives & Museums



Making cultural heritage collections usable as data: challenges, limitations and opportunities

Thank you!

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Inspiration exists, but it has to find you working. Pablo Picasso



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